



Tran Thanh Quan

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SUMMARY

Software Engineering student aspiring to become a **well-rounded Software Engineer**, with an interest in contributing across the software development lifecycle, from design and implementation to testing, debugging, and system improvement. Also interested in exploring **AI and DevOps** through side projects and self-directed learning, including the use of **AI-assisted workflows** in software development. A collaborative team player seeking a **Software Engineering Internship** to further develop practical engineering skills in a professional environment.

EDUCATION

FPT University

Bachelor of Engineering in Software Engineering – GPA: 8.0/10

Ho Chi Minh City, Vietnam

Aug. 2023 – Aug. 2027

EXPERIENCE AND PROJECTS

Vijaco — Internal Document Management App [Vijaco Demo](#) | *Freelance (Solo Developer)* Jun. 2026 – Present

- Independently designed and built an internal document management system end-to-end for **30+ staff**, implement **role- and resource-level permissions** with full audit history to prevent accidental sharing, edits, or deletions.
- Standardized **document lifecycle management** through versioning, retention, and centralized administration, leveraging existing company infrastructure to **reduce deployment and storage costs**.

Hospital Management System [GitHub Link](#) | *Java 17, PostgreSQL | Personal Project* Oct. 2025 – Present

- Developed **118 REST API endpoints** with Spring Boot, JPA, and PostgreSQL across **7 hospital workflows**, using validation, transactions, and locking to maintain data consistency.
- Implemented **Spring Security and JWT** with **34 permissions across 7 roles**, enforcing access boundaries for clinical and administrative operations.

Flash Sale System & Reliability Lab [GitHub](#) | *Java 21, Redis, JMeter | Personal Project* Feb. 2026 – Present

- Built a **Redis Lua stock gate** with compensation and scheduled Redis-MySQL reconciliation to prevent oversell and repair partial-failure drift; validated **5,000 requests across 100 threads** with **0 oversells and drift**.
- Implemented a **Transactional Outbox** with retries and Prometheus monitoring, atomically persisting orders and events before Kafka publishing; removed the dual-write failure window and enabled **at-least-once delivery**.

AI Hospital Knowledge Assistant [GitHub Link](#) | *FastAPI, RAG | Personal Project* Feb. 2026 – Present

- Applied **Graph RAG** to link patient, diagnosis, medication, lab, and encounter data into a visual longitudinal graph, enabling **multi-hop relationship tracing** and richer context for LLM responses.
- Built an **OCR-to-RAG workflow** that converts scanned medical records into searchable evidence for clinical chat; achieved a **95.2% citation score** with **0 unauthorized chunks** across 300 cases.

ACTIVITIES & CERTIFICATIONS

Tet Luon Vui – Community Fundraising Project | *Communications & Logistics* [Evidence](#) | 9 Members | 2026

- Contributed to a fundraising campaign that donated **VND 3.55 million** to the Vietnam Fatherland Front.

Certifications: [Architecting Solutions on AWS \(2026\)](#); [Retrieval Augmented Generation – RAG \(2026\)](#)

Global Consumer Intelligence Data Science Program (The University of Tokyo, 2026)

TECHNICAL SKILLS

Languages: Java, Python, SQL, TypeScript

Backend: Spring Boot, Spring Security, FastAPI, JPA/Hibernate, REST APIs, Redis/RQ

Data & Messaging: PostgreSQL, MySQL, Redis, pgvector, Kafka, Flyway

Testing & Observability: JUnit, Testcontainers, JMeter, k6, Prometheus, Playwright, Grafana, Loki, Elk

DevOps & Cloud: Docker, Docker Compose, Dokploy Traefik, Nginx, Git, GitHub Actions, AWS

Code Quality & Security: AI-assisted PR Review, SonarCloud, CodeQL, Trivy, Dependabot